Standard Bank

AN EXPLAINABLE MACHINE LEARNING-BASED FRAUD DETECTION SYSTEM

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Contents

Problem Statement

The rapid expansion of financial institutions alongside the ubiquitous growth of web-based e-commerce has led to an unprecedented surge in digital transaction volumes (Enaifoghe, 2026). However, this rapid digitization of financial services has also expanded the attack surface for bad actors, making online banking fraud a persistent and rapidly evolving threat (Nanduri et al., 2020). As credit card payment adoption increases, traditional defense mechanisms struggle to keep pace with the increasingly sophisticated and adaptive tactics employed by modern fraudsters, who continually refine their methods to disguise illicit activity as legitimate consumer behavior.

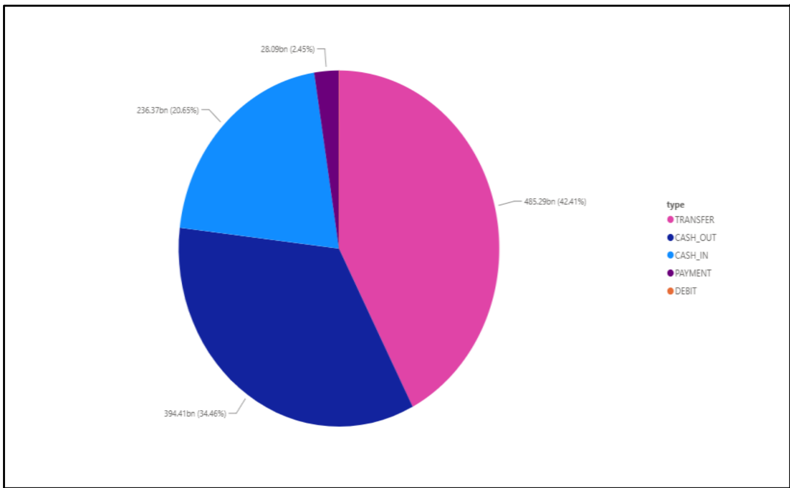
A central challenge in mitigating financial fraud is the iterative nature of modern attack strategies. Fraudsters actively analyze the operational logic of automated detection systems, probing for vulnerabilities and systematically refining their evasion tactics. This constant adaptation renders legacy rule-based and static detection models progressively less effective, creating a complex arms race between security systems and fraudulent entities. Consequently, relying on static or traditional detection frameworks leaves institutions exposed to substantial financial losses and compromises consumer trust, underscoring the critical need to explore and evaluate novel methodologies that significantly enhance the predictive performance and adaptability of existing fraud detection architectures.

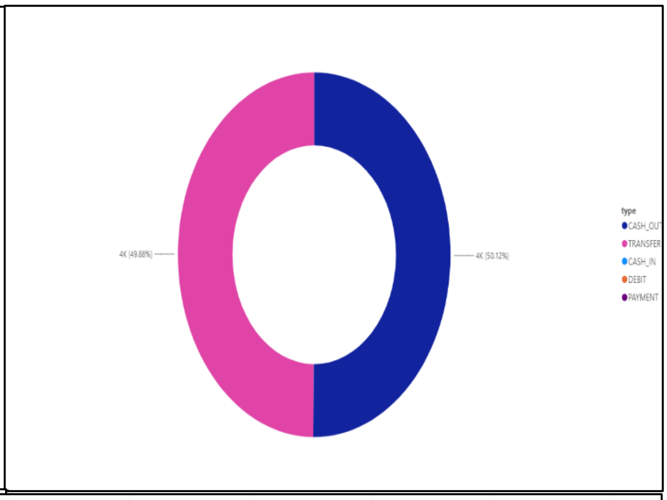
Data Understanding

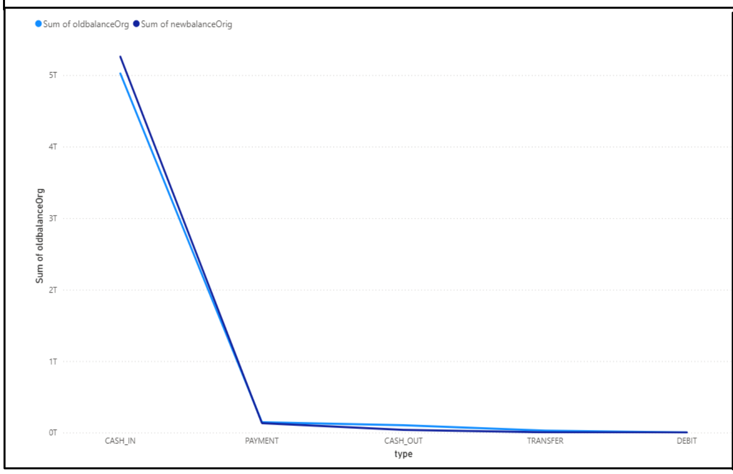
The PaySim dataset is a synthetic financial log generated via Multi-Agent Based Simulation to model mobile money transactions while overcoming the privacy restrictions associated with real-world financial data. Scaled from an actual month of transactional logs from an African mobile money provider, the dataset consists of over 6.36 million records spanning a 30-day period (744 temporal steps, where each step represents one hour). Each transaction record captures key operational attributes, including the transaction type (CASH-IN, CASH-OUT, DEBIT, PAYMENT, and TRANSFER), the transaction amount in local currency, and the unique identifiers for origin (nameOrig) and destination (nameDest) accounts. Crucially, it tracks pre- and post-transaction account balances for both parties (oldbalanceOrg, newbalanceOrig, oldbalanceDest, and newbalanceDest). The target variable, isFraud, flags illicit transactions—specifically patterns where malicious actors attempt account takeovers to empty balances through transfers and immediate cash-outs—while isFlaggedFraud highlights business-rule violations, such as unauthorized single transfers exceeding 200,000 units. The dataset is heavily class-imbalanced, with fraudulent events accounting for less than 0.2% of total transactions and occurring exclusively within the TRANSFER and CASH-OUT types.

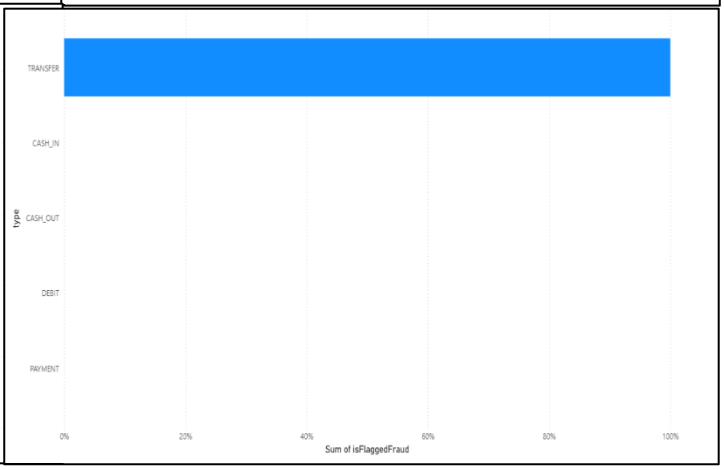
Exploratory Data Analysis

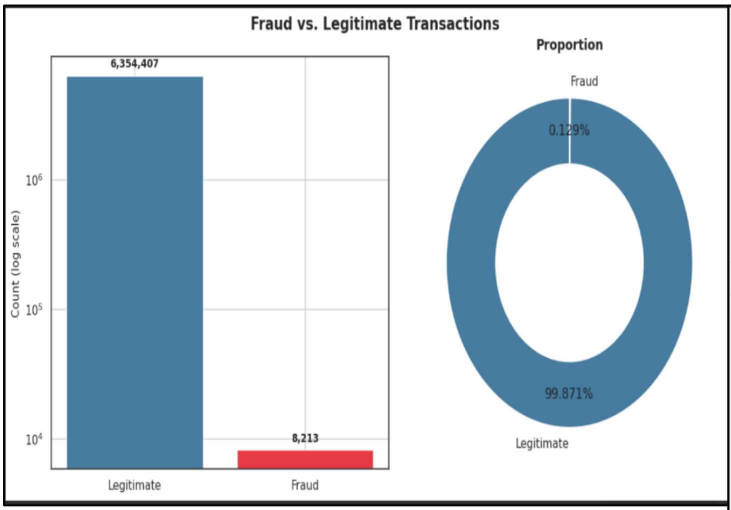
To uncover underlying patterns, detect anomalies, and validate key assumptions before modeling, this Exploratory Data Analysis (EDA) section examines the distributions, relationships among features, and class imbalances inherent in the PaySim financial transaction dataset.

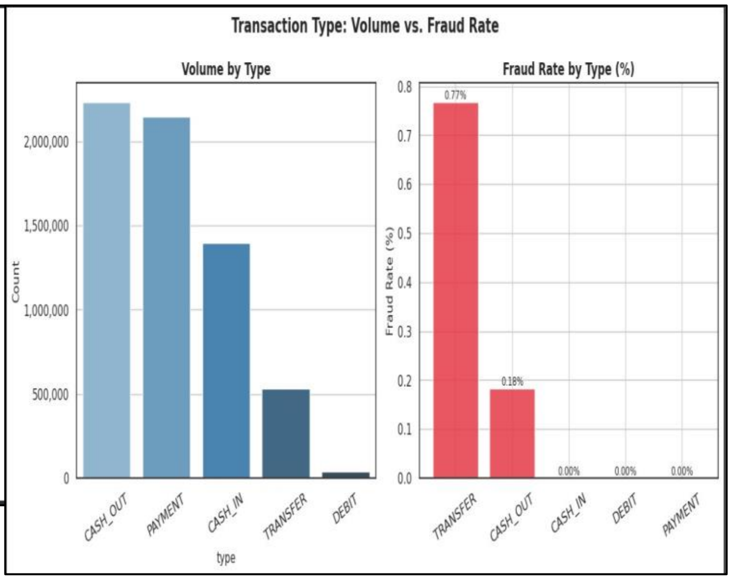






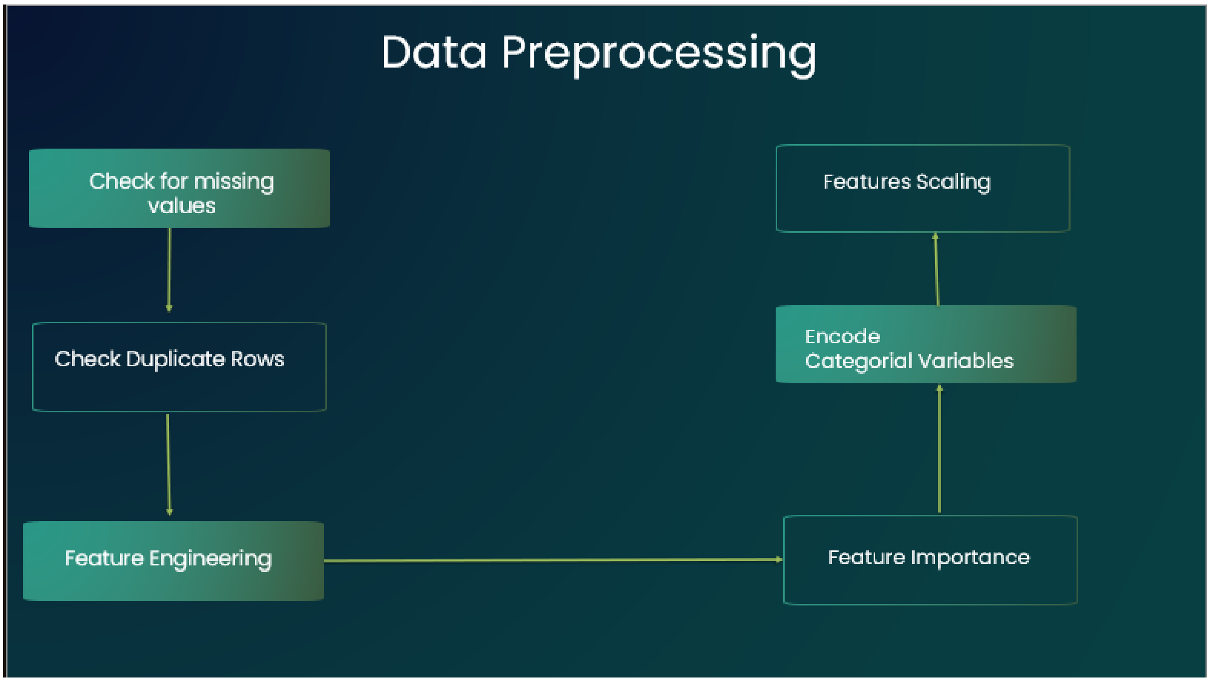






Data Preprocessing

To ensure data quality and prepare the dataset for model training, the following preprocessing steps handle missing values, normalize numerical features, and encode categorical variables.



Model Development and Selection

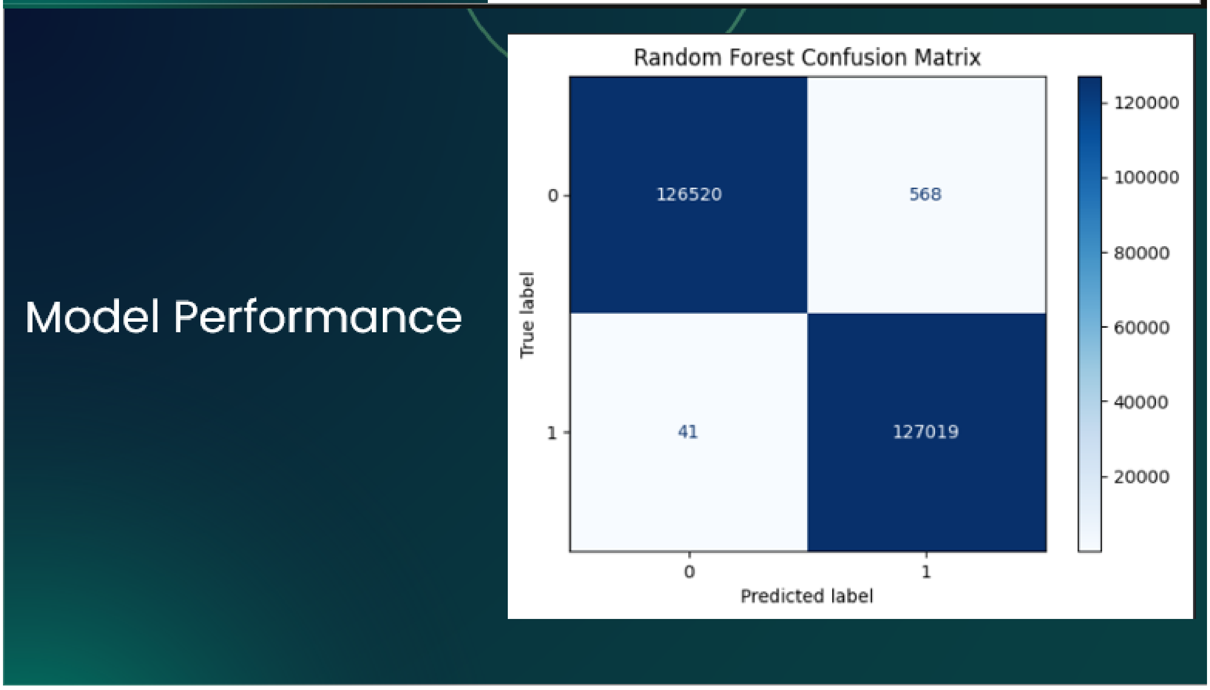
To identify the optimal approach for this task, this section details the model selection criteria, the training process, and the comparative evaluation of candidate algorithms.

Model Performance

This section presents the performance metrics and empirical results used to evaluate, compare, and validate the effectiveness of the trained models

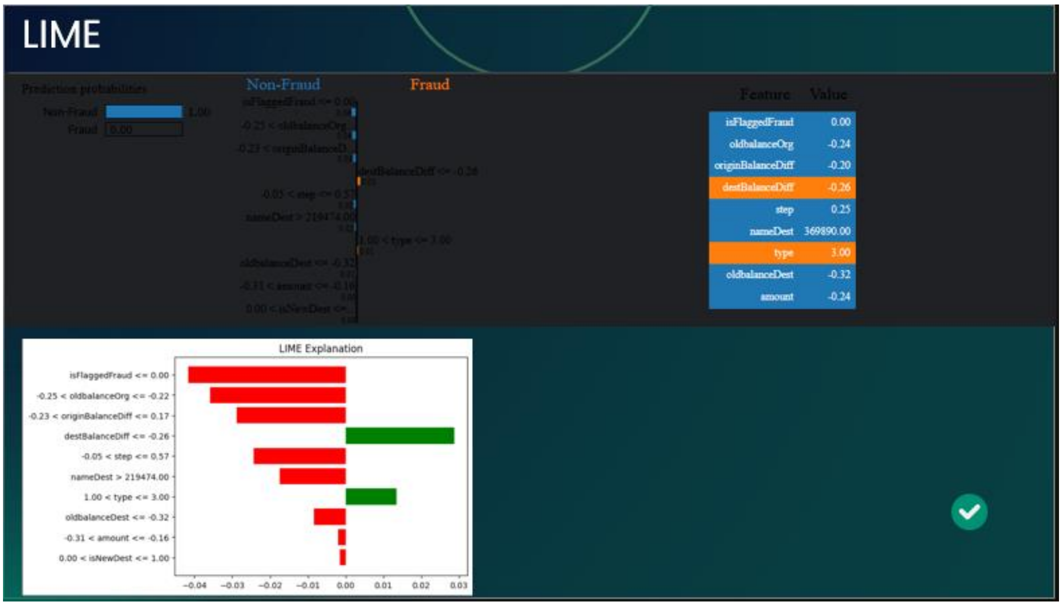
Table 1. Performance Comparison of Machine Learning Models

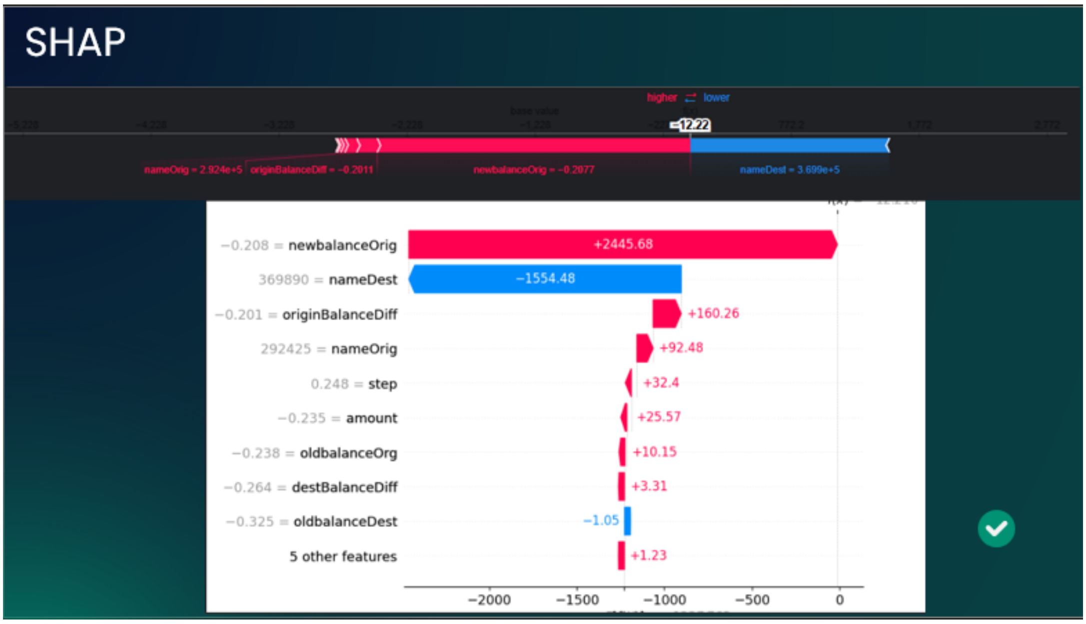
Rank	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
1	Random Forest	0.9992	0.9986	0.9998	0.9992	1.0000
2	CatBoost	0.9976	0.9955	0.9997	0.9976	0.9999
3	XGBoost	0.9985	0.9972	0.9998	0.9985	0.9999
4	LightGBM	0.9985	0.9974	0.9996	0.9985	0.9994
5	Logistic Regression	0.9476	0.9485	0.9466	0.9475	0.9849



Explainable AI

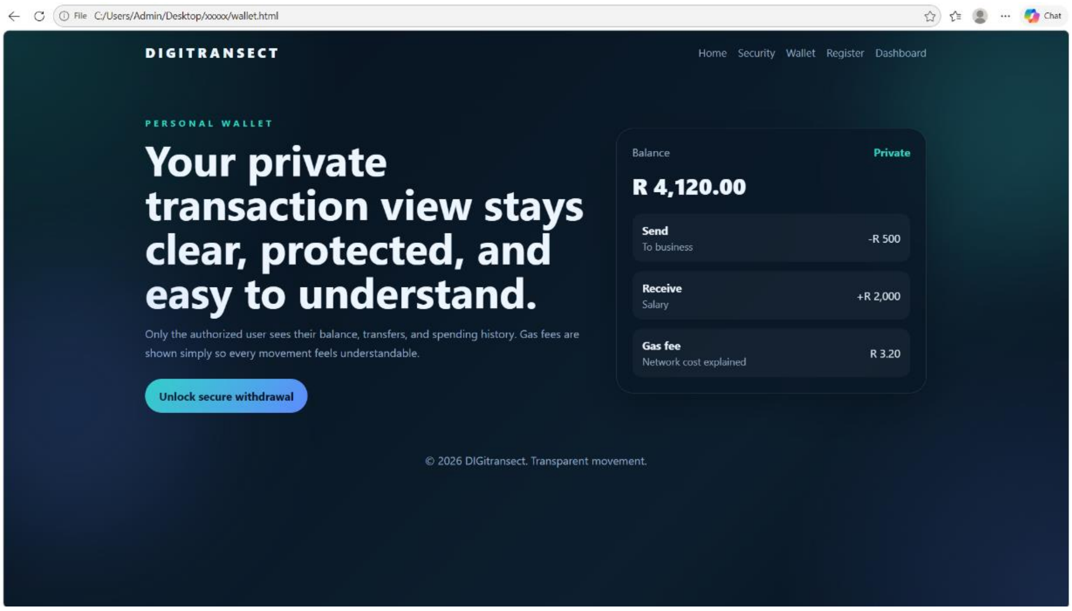
To ensure transparency and trust in the model's predictions, this section applies Explainable AI (XAI) techniques to interpret decision-making processes and highlight key feature contributions.





Model Deployment and System Creation

To address the limitations of existing approaches, this section presents the proposed solution, outlining its core architecture, methodology, and key innovations.



Conclusion

The project successfully processed more than 6.3 million mobile money transactions from the PaySim dataset, with no missing, duplicate, or corrupted records detected during the data quality assessment. This demonstrates that the dataset is well-structured and suitable for large-scale machine learning applications. Initial exploratory data analysis also provided valuable insights into transaction behavior, including the distribution of transaction types, transaction amounts, and temporal patterns across the 30-day simulation period. Establishing a high-quality and reliable dataset forms a strong foundation for developing an accurate and robust fraud detection system.

A key finding of the analysis is the extreme class imbalance between legitimate and fraudulent transactions, with fraudulent activities representing only a very small fraction of the overall dataset. This imbalance presents a significant challenge because conventional machine learning algorithms tend to favor the majority class, resulting in poor detection of rare fraud cases. Consequently, the next phase of the project will focus on implementing appropriate data balancing techniques, such as hybrid over-sampling and under-sampling methods, together with selecting and optimizing advanced machine learning algorithms capable of learning from highly imbalanced data. The effectiveness of these models will be evaluated using metrics such as precision, recall, F1-score, ROC-AUC, and Precision-Recall AUC to ensure that fraudulent transactions are detected accurately while minimizing false alarms. The ultimate objective is to develop a reliable fraud detection system that can identify suspicious mobile money transactions in real time and support more secure digital financial services.

Declaration of the use of Artificial Intelligence tools

ChatGPT was used for coding.

References

- Enaifoghe, A. (2026). The Rapid Growth of Digital Banking Services on Customer Behaviour and Satisfaction in South Africa. Intechopen.com. [online] doi:10.5772/intechopen.1012046.
- J. Nanduri, Y.-W. Liu, K. Yang, and Y. Jia, "Ecommerce fraud detection through fraud islands and multi-layer machine learning model,'' in Proc. Future Inf. Commun. Conf., in Advances in Information and Communication. San Francisco, CA, USA: Springer, 2020, pp. 556–570.